

SQL Agent

A methodology for shipping multi-model LLM systems

Team

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THE PREMISE

An analyst with a CSV.

A question they can't quite write in SQL.

What if the model wrote it?



THE METHODOLOGY

Six phases — from raw datasets to a live agent

Each phase has a question we had to answer. Each one constrained the next. **This is the methodology we'd use again on any LLM project.**

Don't reinvent — curate from what already exists

Ten public text-to-SQL datasets, merged into one canonical training set:

- `b-mc2/sql-create-context`
- `gretelai/synthetic_text_to_sql`
- `knowrohit07/know_sql`
- `NumbersStation/NSText2SQL`
- `Clinton/Text-to-sql-v1`
- `motherduckdb/duckdb-text2sql-25k`
- `bugdaryan/spider-natsql-wikisql`
- `ChrisHayduk/Llama-2-SQL`
- `kaxap/llama2-sql-instruct`
- `PipableAI/spider-bird`

The question we faced:

"Should we build our own dataset, or assemble from public sources?"

Building from scratch would have taken weeks and produced a smaller, lower-quality corpus. **The internet had already done most of the work.**

Combined raw size: **~1.2 million rows**. Combined quality: *uneven*. Schemas inconsistent. Duplicates plentiful.

From 1.2M raw rows to 723k usable training examples

1.2M

RAW

761k

DEDUPLICATED

93%

PASS LENGTH FILTER

723k

FINAL TRAINING SET

All transforms reproducible via UV scripts in `training/data_pipelines/`.

One dataset wasn't enough — we built three

DATASET · 01

text-to-sql-mix-v2

761k

NL → SQL pairs from 10 merged public sources

DATASET · 02

chart-reasoning-mix-v1

75k

nvBench (25k) + GPT-4.1-nano knowledge distillation (50k)

DATASET · 03

svg-chart-render-v1

25k

nvBench charts re-rendered through matplotlib's SVG backend

Three roles. Three datasets. Each one published openly on Hugging Face under permissive licenses.

text-to-sql-mix-v2	huggingface.co/.../text-to-sql-mix-v2
chart-reasoning-mix-v1	huggingface.co/.../chart-reasoning-mix-v1
svg-chart-render-v1	huggingface.co/.../svg-chart-render-v1

Fine-tune a 7B model on a single GPU — without paying \$1,000

The decision point:

"Vanilla transformers on a 4× A100 cluster, or Unsloth on one L40S?"

Unsloth uses 4-bit quantization + LoRA + custom Triton kernels. Same training math, dramatically less memory and time.

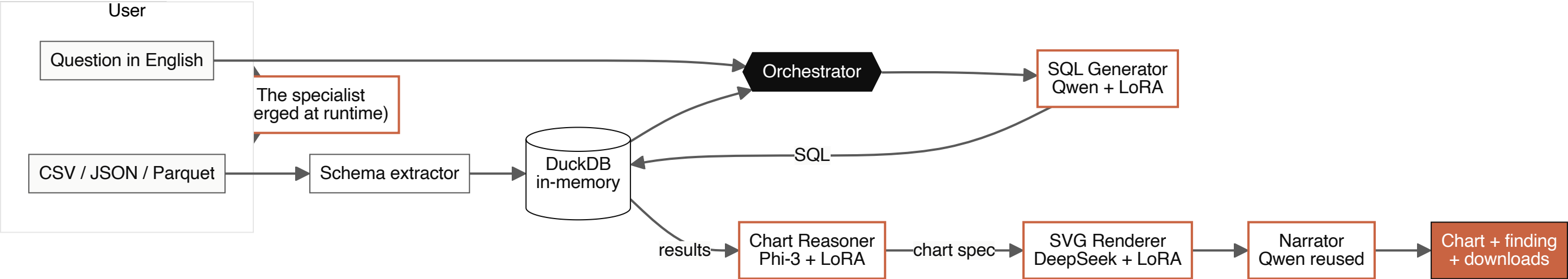
	VANILLA	UNSLOTH
7B on 48 GB GPU	won't fit	fits
Speed	1×	2×
Memory	1×	40% less
Output	14 GB	160 MB adapter

13.5 hours, one L40S, one epoch, loss 0.27

Base	Qwen2.5-Coder-7B-Instruct
Method	QLoRA r=16, α=32 (4-bit)
Examples used	672,949
Sequences after packing	154,462
Hardware	1× NVIDIA L40S (48 GB)
Wall-clock	13.5 hours
Final loss	0.2658
Total cost	~\$24

The trick that mattered: `packing=True` in TRL. Concatenated 723k examples into 154k sequences of 1024 tokens. **4× speedup. ~\$14 saved.**

What we ship is the diff, not the model



ADAPTER	BASE	ADAPTER SIZE
sql-generator-qwen25-coder-7b-lora	Qwen 2.5 Coder 7B	161 MB
chart-reasoner-phi3-mini-adapter-only	Phi-3 Mini 4k	38 MB
svg-renderer-deepseek-coder-1.3b-lora	DeepSeek Coder 1.3B	22 MB

Combined adapter weight: **221 MB**. Base models stay public — we just teach them new tricks.

Per query: 4 LLM calls, 5–8 seconds, end to end

Three adapters loaded **once at module level** into a half-H200 via Hugging Face's ZeroGPU. Self-correcting SQL: if a query fails, the model retries with the error in context.

\$30. All in.

STAGE	COMPUTE	COST
SQL Generator training	HF Jobs L40S, 13.5 h	~\$24
Chart Reasoner training	Colab / HF Jobs	~\$3
SVG Renderer training	Colab / HF Jobs	~\$1
Chart dataset distillation	gpt-4.1-nano Batch	~\$2.50
Inference hosting	HF Spaces ZeroGPU	\$0
Total		~\$30

\$30

ALL-IN TO TRAIN THREE PRODUCTION FINE-TUNES

Less than a steak dinner.
Less than one OpenAI eval.
Open weights, reproducible.

LIVE DEMO

Drop a CSV.

Type a question.

Three fine-tuned models answer in five seconds.

— huggingface.co/spaces/DanielRegaladoCardoso/sql-agent

CLOSING

Three fine-tunes. Thirty dollars. Thirteen and a half hours.

Open source

- Repo · github.com/DanielRegaladoUMiami/sql-agent-llmops
- Space · hf.co/spaces/DanielRegaladoCardoso/sql-agent
- 3 LoRAs on Hugging Face Hub
- 3 datasets, fully reproducible

Where it goes next

- Multi-turn conversation memory
- Eval on Spider / WikiSQL / BIRD
- Anomaly detection at upload
- Statistical summary at ingest

Thank you.

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